George Zhang

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Canadian & Hong Kong citizen · eligible for U.S. internships (J-1); TN-eligible post-graduation

Education

University of Waterloo Waterloo, ON

Bachelor of Computer Science (Honours, Co-op) *Sep 2024 – May 2029*

- Relevant coursework: Data Structures & Algorithms, OOP (C++), Computer Organization, Logic & Computation, Optimization, Probability & Statistics.

Experience

Ford Motor Company Waterloo, ON

Software Developer Intern — Android Automotive OS *May 2026 – Present*

- Develop and ship features for the in-vehicle Car Dialer app on Android Automotive OS (AAOS), part of Ford's production infotainment platform, across a multi-module Java/Kotlin codebase.
- Build call-management flows on the Android Telecom framework with Bluetooth Hands-Free Profile (HFP) integration, validated by Robolectric unit tests and delivered through a Gradle / AOSP-Soong CI pipeline.

University of British Columbia Vancouver, BC

Undergraduate Research Assistant — Centre for Brain Health *Jun 2025 – Aug 2025*

- Led development of two full-stack hand-rehabilitation apps within PIKA, an AI Parkinson's-care platform deployed at Vancouver Coastal Health.
- Built a real-time computer-vision pipeline (Python, OpenCV, MediaPipe) detecting 7 hand-exercise gestures at ~90% accuracy with under 100 ms latency, fully on-device to meet PIPEDA/GDPR requirements, plus activity-heatmap dashboards that cut clinician monitoring overhead by 30%.
- Presented the work at the 11th Singapore International Parkinson Disease & Movement Disorders Symposium.

Projects

Interactive Portfolio & Game Engine | TypeScript, Next.js, React, Canvas, Redis georgezhang.ca

- Built a personal site with two faces — a fast web portfolio (custom cursor + water-ripple canvas) and an explorable pixel-art "museum" on a custom HTML5 Canvas engine (60 fps loop, AABB collision, A* pathfinding, particles, procedural maps), with multiplayer "ghost trails" (Upstash Redis) and a daily GitHub Actions content pipeline.

Code-Aware RAG System | Python, tree-sitter, embeddings github.com/TheYellowDuck/RAG-codebase

- Engineered a retrieval-augmented-generation system answering codebase questions with citation-backed answers, chunking on AST boundaries (tree-sitter) and fusing dense + BM25 retrieval (RRF) with a code graph and personalized PageRank.
- Treated each design choice as a measured ablation — the LLM reranker significantly improved ranking (MRR/NDCG, $p < 0.001$) while a cross-encoder was net-negative on code — with bootstrap CIs and significance tests; provider-agnostic (Anthropic/OpenAI/local).

Limit Order Book & Matching Engine | C++20 github.com/TheYellowDuck/limit-order-book

- Architected a low-latency, lock-free matching engine with strict price-time priority and 9 order types (limit, market, IOC, FOK, stop, iceberg, pegged) on custom cache-optimized data structures.
- Engineered a wait-free runtime (SPSC ring buffer, single-writer matcher, seqlock market-data publishing) with deterministic journaling/replay; validated with differential, property-based, and fuzz testing under sanitizers.

Autonomous LLM Web Agent | Python, Playwright github.com/TheYellowDuck/web-agent

- Designed a ReAct + reflection agent over a browser accessibility tree with a WebArena / Online-Mind2Web eval harness; showed the same agent scores 0.43 vs 0.61 by scorer alone, isolating measurement from real capability.

Technical Skills

Languages	C++, Python, Java, Kotlin, TypeScript/JavaScript, SQL, C, HTML/CSS
Frameworks	React, Next.js, PyTorch, OpenCV, MediaPipe, NumPy, Pandas, Android SDK/Jetpack, Tailwind
Tools	Git, Linux, Docker, Gradle, GitHub Actions, Redis, Playwright, Android Studio
Concepts	Data Structures & Algorithms, Concurrency, OOP & Design Patterns, LLMs / RAG, Computer Vision

Awards & Competitive Programming

- President's Scholarship of Distinction & Entrance Scholarship in Mathematics, University of Waterloo
- ICPC Local (Waterloo) — Rank 28 / ~100 · 470+ LeetCode & 220+ DMOJ solved (1,497 points)
- Canadian Computing Competition — Junior: perfect score, Senior: top quartile · AIME Qualifier · COMC Distinction
- RoboCup International Junior Maze Rescue 2023 (Captain) — 2nd Place Supergroup, 17th Individual